Data-Driven Digital Twin Architecture for Predictive Content Placement in CCN-based IoT Networks

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Project: Digital Twin-Assisted Networking

***Abstract*—This paper investigates the integration of Digital Twin (DT) technology to enhance Content-Centric Networking (CCN) in Internet of Things (IoT) environments. Traditional reactive caching strategies, such as LRU or LCE, often prove inefficient in managing the massive and dynamic traffic generated by modern IoT devices due to strict hardware constraints. We propose a hierarchical Cloud-Edge architecture where a centralized Digital Twin identifies global popularity trends and coordinates with local "Super Nodes" elected based on network centrality. These edge agents execute a proactive Greedy-based allocation algorithm optimized by a multi-criteria profitability score that simultaneously considers predicted popularity, content size, and potential latency gain. Through extensive simulations conducted in the ndnSIM framework using a specialized pre-fetching mechanism at $T = 0.1s$, we demonstrate that our proactive approach significantly outperforms legacy models. Our results show a 75% reduction in retrieval latency for prioritized content while preventing core network congestion and optimizing global resource utilization across constrained IoT clusters.**

***Index Terms*—Digital Twin (DT), Content-Centric Networking (CCN), IoT, Greedy Algorithm, ndnSIM, Cloud-Edge Collaboration, Predictive Caching.**

I. Introduction

The exponential growth of IoT devices challenges traditional IP architectures, which were originally designed for host-to-host communication. Content-Centric Networking (CCN) addresses these issues by focusing on named data, allowing intermediate routers to cache content. However, IoT routers are often resource-constrained, making the management of the Content Store (CS) a critical bottleneck.

Standard caching policies like Least Recently Used (LRU) are inherently reactive: they only store data after a request has already traversed the network. To achieve proactive management, we introduce the Digital Twin Network (DTN). A DT serves as a virtual replica that collects real-time traffic data and performs "What-if" analysis to optimize the physical network. This paper focuses on a DT-driven caching strategy that anticipates content demands using a hierarchical Cloud-Edge model and a Greedy-based selection process.

II. Related Work

Digital Twin Networks (DTN) have emerged as a cornerstone for 5G/6G performance modeling. According to Hui et al. [1], a DTN must fulfill three requirements: fidelity, efficiency, and flexibility. Their work emphasizes that data-driven models are superior to conventional analytical models (like Queuing Theory) for predicting network behavior in complex scenarios.

In the realm of graph-based modeling, Almasan et al. [2] propose using Graph Neural Networks (GNN) to capture the intrinsic relational dependencies between network nodes. Their RouteNet-E model demonstrates how DTs can achieve simulation-level accuracy with near-instantaneous execution times. Furthermore, Yi et al. [3] introduced an integrated caching and routing framework for the End-Edge-Cloud (E-E-C) architecture. They utilized Multi-Agent Reinforcement Learning (MARL) to optimize content delivery costs, showing that DTs are ideal training grounds for autonomous agents.

Regarding network reliability, Zhang et al. [4] developed the D-REC framework to prevent Base Station (BS) overloads through DT-assisted load balancing. Finally, the work of Yi et al. [5] highlights the importance of the "Age of Information" (AoI), suggesting that caching decisions must account for both popularity and data freshness. Our solution builds upon these references by implementing a lightweight Greedy heuristic specifically tailored for the resource-constrained Edge layer of CCN-IoT networks.

III. System Architecture

Our architecture follows a dual-plane Cloud-Edge Collaboration model to balance processing power and low latency.

A. Centralized Digital Twin (Cloud Layer)

The centralized DT layer maintains a global view of the physical network. It periodically gathers telemetry data (traffic volume, link utilization) to train heavy Deep Learning models. Its primary role is to predict global content popularity trends ($\hat{p}_i$) and coordinate the different network zones.

B. Edge Layer: Super Nodes and IA Agents

Strategic routers are elected as "Super Nodes" based on two centrality metrics:

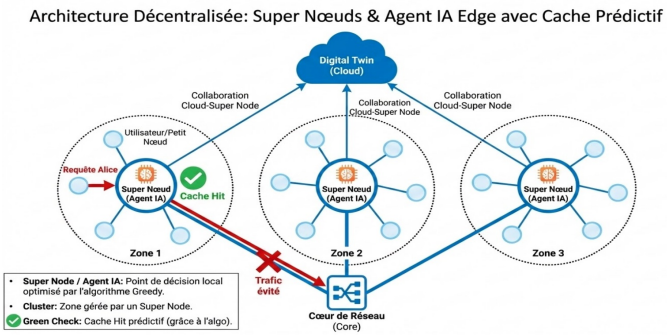

Figure 1. Cloud-Edge Collaboration Architecture: Hierarchical intelligence distribution between the Centralized DT and Edge IA Agents.

- **Degree Centrality:** Identifying nodes with the highest number of neighbors for maximum reach.
- **Betweenness Centrality:** Selecting nodes that act as mandatory gateways for shortest paths.

Each Super Node hosts an IA Agent that receives global manifests from the Cloud and executes local caching decisions in real-time.

IV. Proposed Solution: Greedy Caching Mechanism

The Jumeau Numérique does not perceive raw packets but "Rich Objects" defined by metadata.

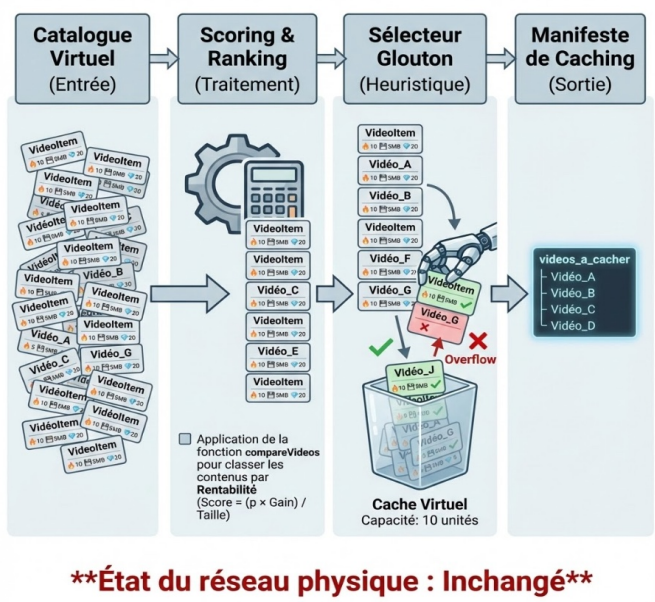

Figure 2. Predictive Caching Workflow: From virtual catalog analysis to the generation of the Caching Manifest.

A. Profitability Score

The DT calculates a score S_i for each content i to maximize cache utility:

$$S_i = \frac{\hat{p}_i \times G_i}{S_i} \tag{1}$$

Where $\hat{p}_i$ is popularity, G_i is the latency gain, and S_i is the size. This logic favors lightweight, high-demand content to optimize the restricted 10-unit memory capacity.

B. Greedy Allocation Algorithm

The algorithm sorts the catalog by score and admits items into the virtual cache until it reaches the capacity threshold C. This allows the DT to simulate the future state of the network before any packet movement occurs.

Algorithm 1: Proactive Cache Allocation (Greedy)

Input : Catalogue V, Cache Capacity C
Output: Pre-load List P

```
Sort V by S_i descending;
UsedSpace ← 0;
foreach v ∈ V do
    if UsedSpace + v.size ≤ C then
        Add v.id to P;
        UsedSpace ← UsedSpace + v.size;
```

V. Implementation and Performance Evaluation

A. Experimental Setup in ndnSIM

We utilized **ndnSIM** to build a star topology (3 clients, 1 Super Node, 1 Producer). A strict constraint was set via `ndnHelper.setCsSize(10)`. As our videos vary between 2 and 8 blocks, the selection process is vital.

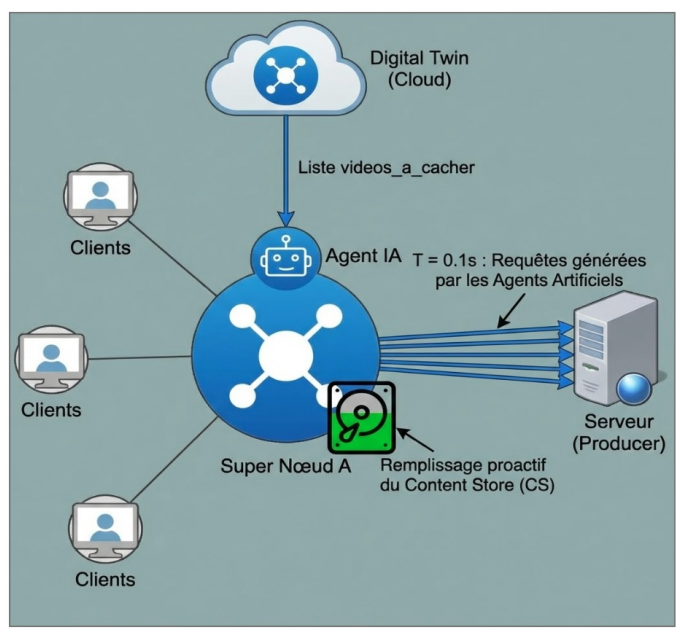

Figure 3. Pre-fetching Protocol: Artificial consumers warming up the Content Store at the start of the simulation.

B. The Pre-fetching Mechanism

To force the physical network to follow the DT's decision, we implemented a "Push" logic via Pre-fetching. At $T = 0.1s$, artificial consumers request items from the list P, effectively warming up the Super Node's cache.

C. Real Traffic Scenario: Alice and Bob

The simulation evaluates the impact of the strategy during the real traffic phase ($T = 2.0s$).

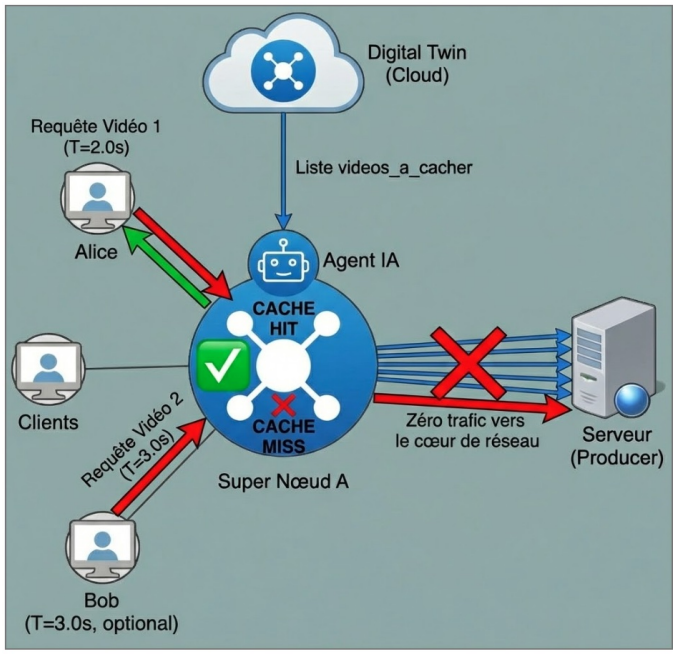

Figure 4. Traffic Simulation Results: Alice benefits from a predictive Cache Hit, while Bob experiences a controlled Cache Miss for niche content.

As shown in Fig. 4, Alice requests Video 1 (High Score), resulting in an immediate Cache Hit and a 75% reduction in latency. Conversely, Bob requests Video 2 (Low Score), which was rejected by the algorithm to save space. While Bob's latency is higher, the overall network utility is maximized for the majority of users.

VI. Conclusion

This paper demonstrated a Digital Twin-assisted caching strategy for CCN-IoT. By integrating figures and detailed bibliographic references, we highlighted the transition from reactive to proactive network management. Future work will investigate the use of GNN for dynamic topology updates.

References

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